

ERGA Assembly Report

v24.10.15

Tags: ERGA-BGE

TxID	3014478
ToLID	imAmeSpa111
Species	Ameles spallanzania
Class	Insecta
Order	Mantodea

Genome Traits	Expected	Observed
Haploid size (bp)	2,583,211,628	2,605,433,968
Haploid Number	8 (source: ancestor)	14
Ploidy	2 (source: ancestor)	2
Sample Sex	NA	XX

EBP metrics summary and curation notes

Obtained EBP quality metric for hap1: 6.8.Q62

Obtained EBP quality metric for hap2: 6.8.Q62

The following metrics were automatically flagged as below EBP recommended standards or different from expected:

- . Observed Haploid Number is different from Expected
- . Observed sex is different from Sample sex

Curator notes

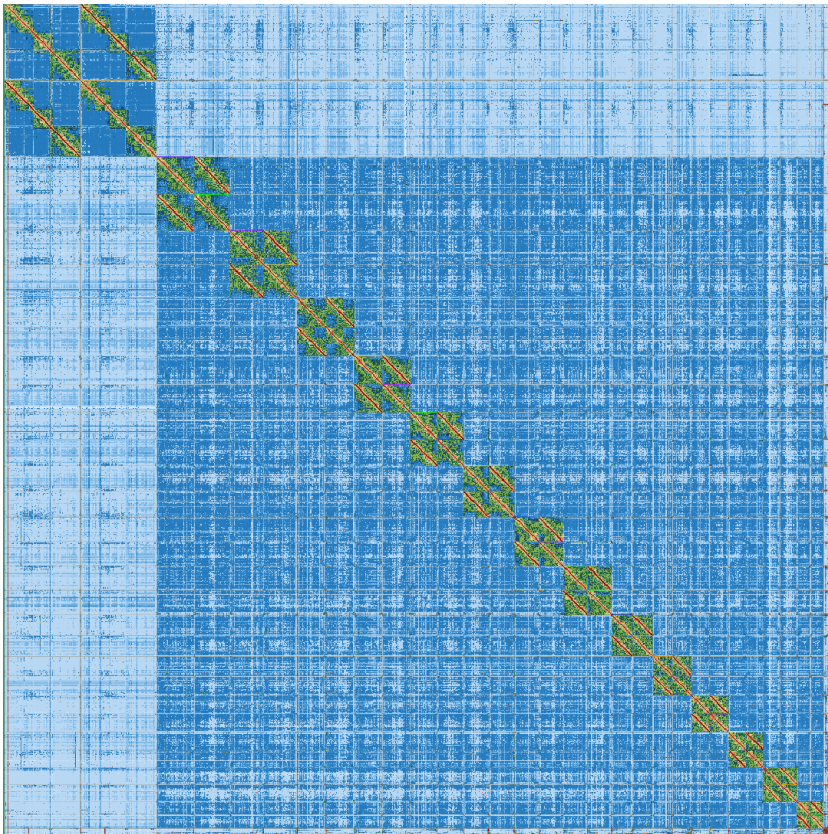
- . Interventions/Gb: None
- . Contamination notes: "Decontamination was done with NCBI FCS pipeline and confirmed with Blobtoolkit"
- . Other observations: "This genome assembly was produced by assembling PacBio HiFi data from two different individuals with Hifiasm, purging with purge_dups and then scaffolding with Yaho. Both assemblies are presented as requested. Curation made 18 cuts in contigs, 14 breaks at gaps and 80 joins for the Hap1 and 17 cuts in contigs, 6 breaks at gaps and 413 joins for the Hap2. The sex chromosomes were assigned based on an alignment with the species *Deroplatys truncata* (GCA_030765065.1)"

Quality metrics table

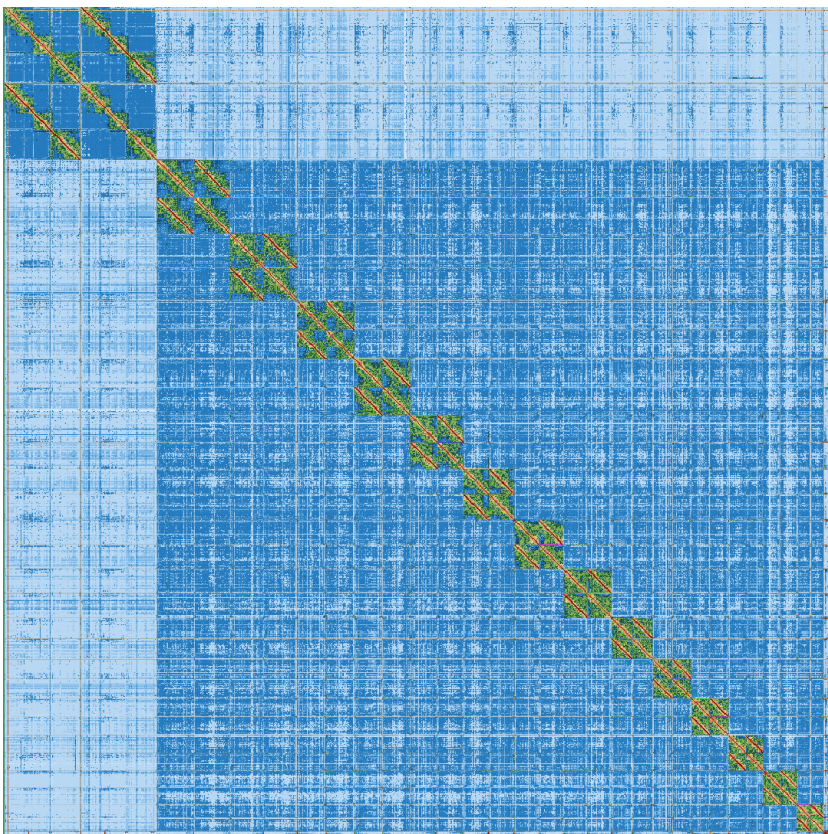
Metrics	Pre-curation hap1	Pre-curation hap2	Curated hap1	Curated hap2
Total bp	2,609,282,592	2,583,173,298	2,605,433,968	2,584,298,143
GC %	37.88	37.88	37.88	37.88
Gaps/Gbp	378.65	341.83	550	510.78
Total gap bp	197,600	176,600	286,600	264,000
Scaffolds	646	819	254	421
Scaffold N50	17,033,530	17,239,675	167,826,513	164,258,888
Scaffold L50	41	38	6	6
Scaffold L90	161	176	13	13
Contigs	1,634	1,702	1,687	1,741
Contig N50	3,996,391	4,133,618	3,681,516	4,022,341
Contig L50	180	159	188	164
Contig L90	757	708	797	742
QV	62.6925	62.6457	62.6942	62.6448
Kmer compl.	94.9093	95.5045	94.8929	95.4991
BUSCO sing.	96.0%	96.2%	95.8%	96.1%
BUSCO dupl.	3.2%	2.7%	3.3%	2.7%
BUSCO frag.	0.1%	0.3%	0.2%	0.3%
BUSCO miss.	0.7%	0.8%	0.7%	0.9%

BUSCO: 5.8.0 (euk_genome_min, miniprot) / Lineage: insecta_odb10 (genomes:75, BUSCOs:1367)

HiC contact map of curated assembly

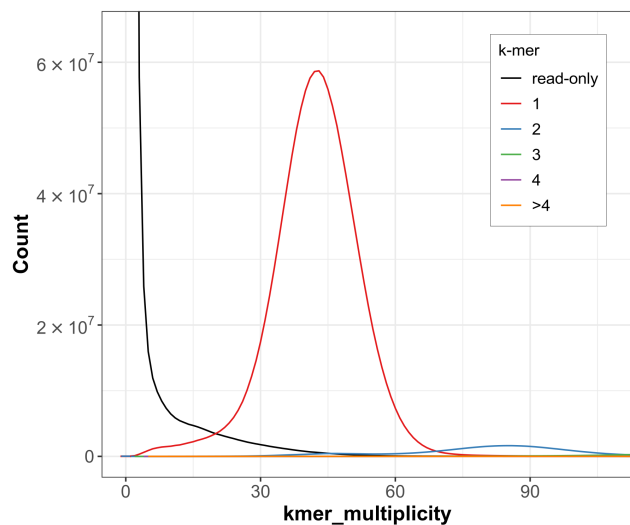


hap1 [\[LINK\]](#)

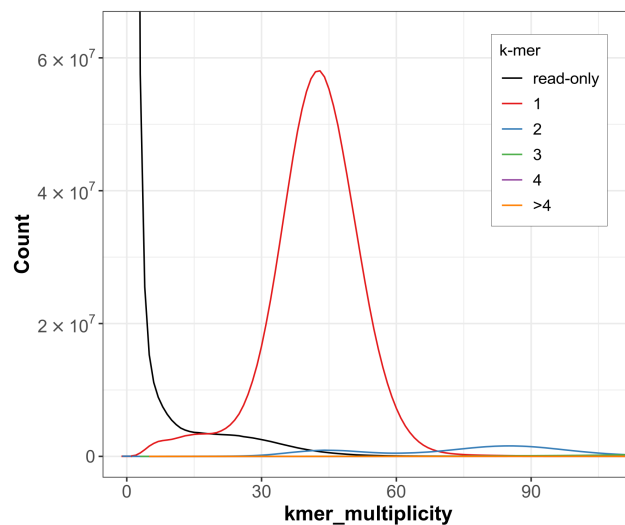


hap2 [\[LINK\]](#)

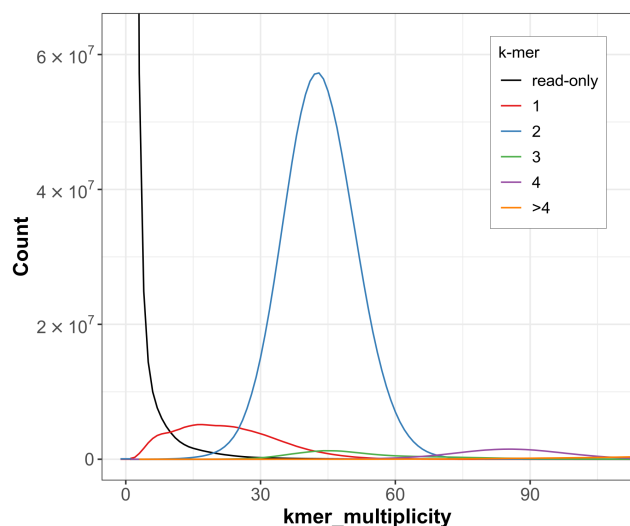
K-mer spectra of curated assembly



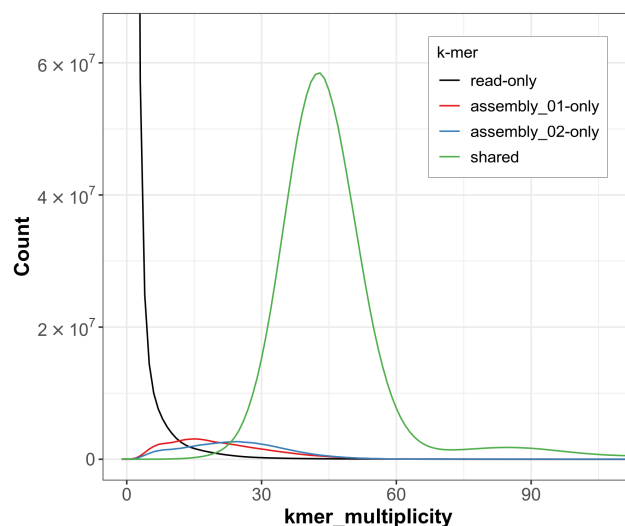
Distribution of k-mer counts per copy numbers found in **assembly_02** (hapl.)



Distribution of k-mer counts per copy numbers found in **assembly_01** (hapl.)

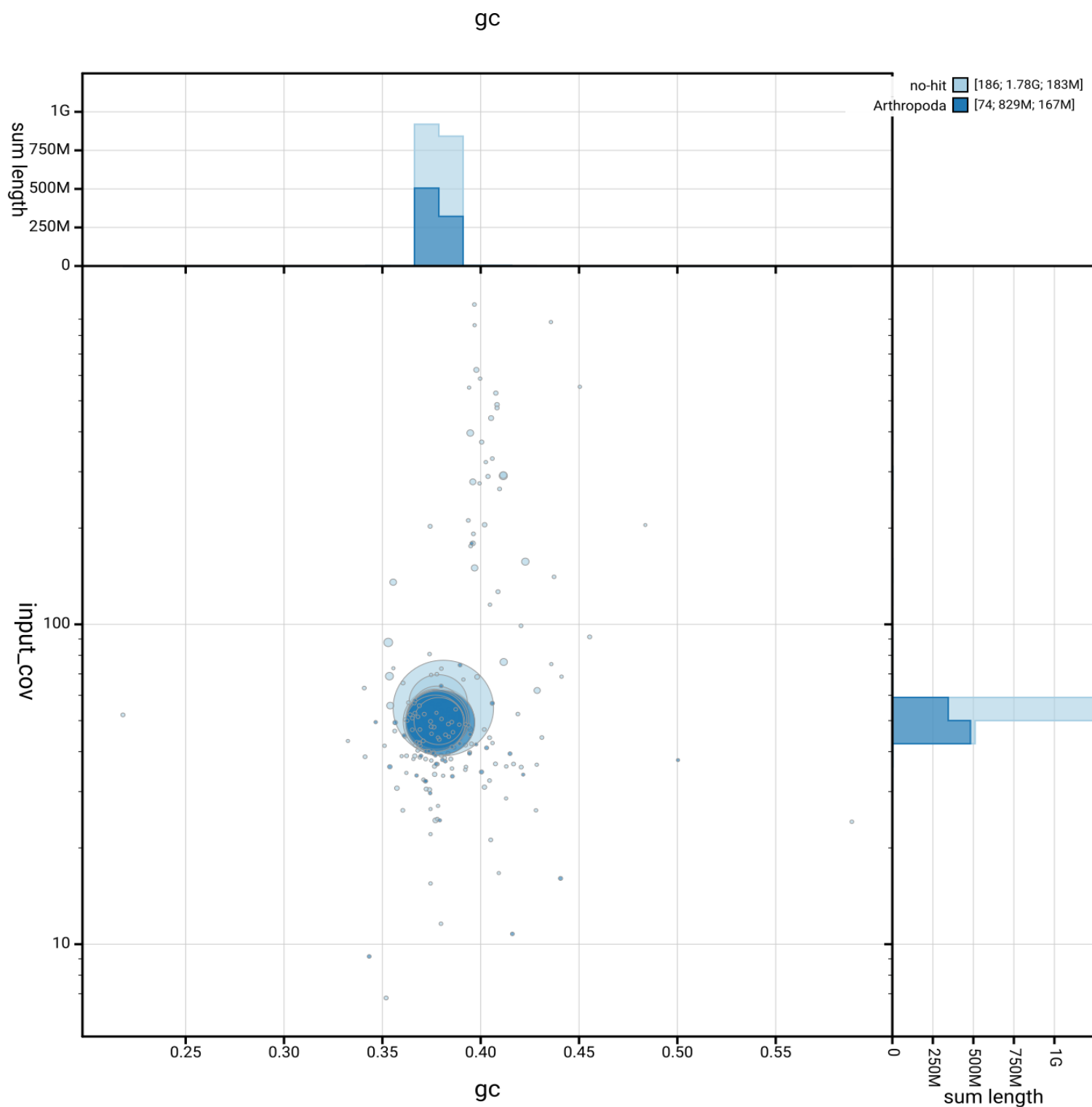


Distribution of k-mer counts per copy numbers found in **asm** (dipl.)

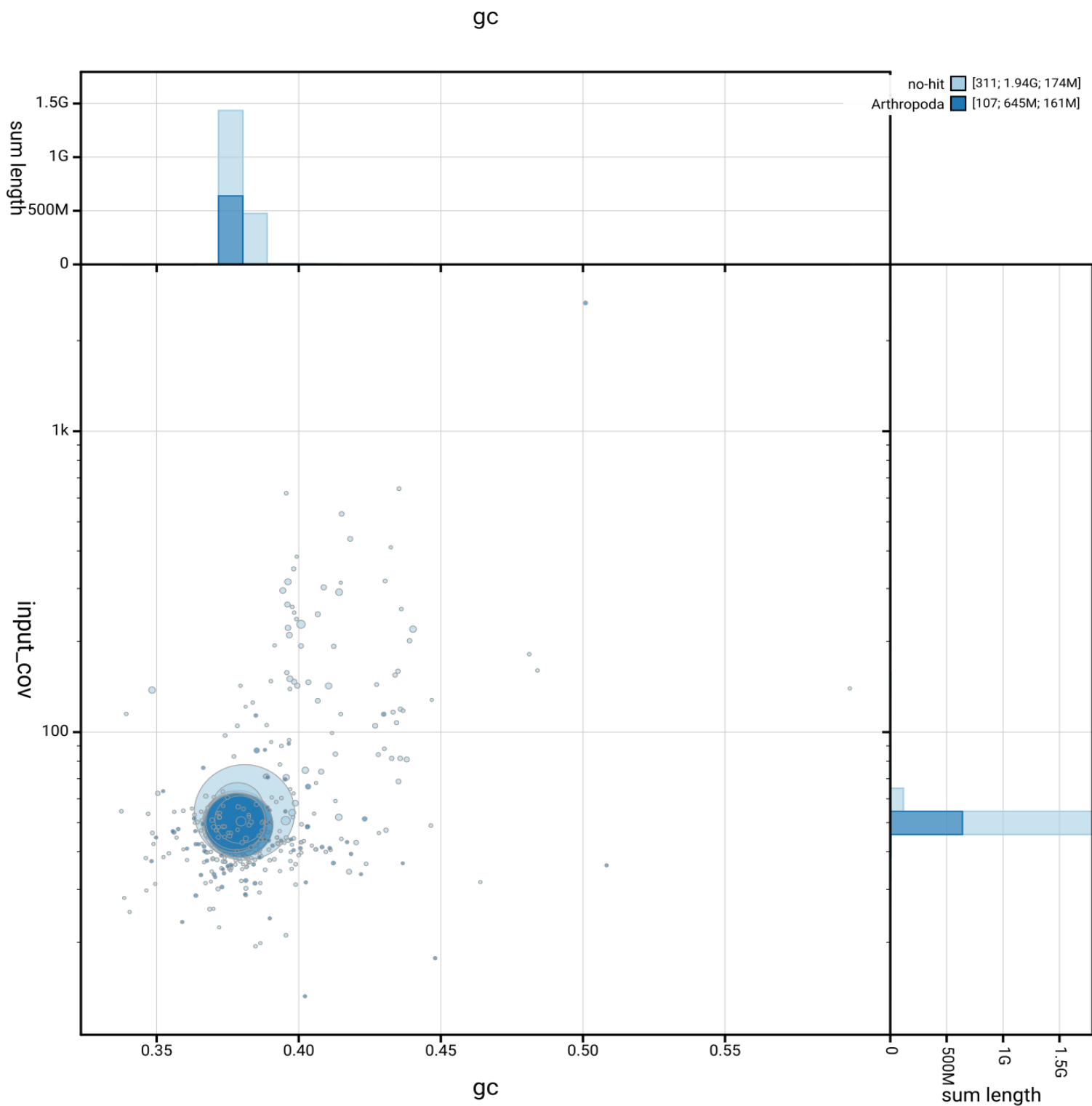


Distribution of k-mer counts coloured by their presence in reads/assemblies

Post-curation contamination screening



hap1. Bubble plot circles are scaled by sequence length, positioned by coverage and GC proportion, and coloured by taxonomy. Histograms show total assembly length distribution on each axis.



hap2. Bubble plot circles are scaled by sequence length, positioned by coverage and GC proportion, and coloured by taxonomy. Histograms show total assembly length distribution on each axis.

Data profile

Data	HiFi	HiC
Coverage	29.54X	145.70X

Assembly pipeline

- **Hifiasm**
 - |_ *ver*: Galaxy Version 0.24.0+galaxy0
 - |_ *key param*: VGP Galaxy workflow VGP3 v 0.3.2
- **purge_dups**
 - |_ *ver*: Galaxy Version 1.2.6+galaxy0
 - |_ *key param*: VGP Galaxy workflow VGP6 v 0.8
- **YaHS**
 - |_ *ver*: Galaxy Version 1.2a.2+galaxy2
 - |_ *key param*: VGP Galaxy workflow VGP8 v 2.0

Curation pipeline

- **PretextMap**
 - |_ *ver*: Galaxy Version 0.1.9+galaxy1
 - |_ *key param*: VGP workflow VGP PretextMap generation v2
- **PretextView**
 - |_ *ver*: 1.0.0
 - |_ *key param*: NA

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